

Member ID: _____

Time: _____

Rank: _____



PYTHON PROGRAMMING-PILOT (355)

NATIONAL – 2022

PRODUCTION PORTION:

Program 1: Character Information _____ (400 points)

TOTAL POINTS _____ (400 points)

Test Time: 90 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

You will have ninety (90) minutes to complete your work.

Your name and/or school name should *not* appear on work you submit for scoring.

1. Create a folder on the flash drive provided using your Member ID as the name of the folder.
2. Copy your entire solution/project into this folder.
3. Submit your entire solution/project so that the graders may open your project to review the source code.
4. Ensure that the files required to run your program are present and will execute on the flash drive provided.

*Note that the flash drive letter may *not* be the same when the program is scored as it was when you created the program.

*It is recommended that you use relative paths rather than absolute paths to ensure that the program will run regardless of the flash drive letter.

The graders will *not* compile or alter your source code to correct for this.
Submissions that do *not* contain source code will *not* be scored.

Assumptions to make when taking this assessment:

- The user will *not* enter invalid input.

Development Standards:

- Your Code must use a consistent variable naming convention.
- All functions (if any) must be documented with comments explaining the purpose of the method, the input parameters (if any), and the output (if any).
- If you create a class, then you must document the class and its methods.

Character Information

You are tasked with writing a program that processes video game character information using classes. The base class (*Character*) keeps data attributes for the following pieces of information:

- Character name
- Character id_number

The program must also have a class named *Hero* that is a subclass of the *Character* class. The *Hero* class should keep data attributes for the following information:

- level (1, 2, 3, 4, 5)
- loot (formatted with two decimal points)

Additionally, as with most games, there will be a boss at the end of each level. Program a *Boss* class that is a subclass of the *Character* class. The *Boss* class should keep data attributes for the following information:

- level (1, 2, 3, 4, 5)
- hp (integer)
- attack_damage (integer) (damage inflicted on hero)
- lifespan (NOTE: the number of attacks the boss can withstand by dividing the hp by 300)

Once you have written the classes, write a program that does the following:

1. Creates an object of the *Hero* class and prompts the user to enter data for Boss A (see Table 1).
2. Creates an object of the *Boss* class and prompts the user to enter data for Boss A (see Table 1).
3. Stores the data in the respective objects.
4. Retrieve and display the data on the screen from the respective objects.

Attribute	Hero A	Boss A
Name	Knight	Ogre
Character ID	111	333
Level	1	2
Loot	50.00	
Hit Points (hp)		5500
Attack Damage		99
Lifespan		

Requirements:

1. You must create a python program called characterinformation.py.
2. You must create the *Character*, *Hero*, and *Boss* classes within characters.py
3. Your program must create the Hero object.
4. Your program must prompt the user to enter in the correct Hero attributes for Hero A.
5. Your program must create the Boss object.
6. Your program must prompt the user to enter in the correct Boss attributes for Boss A.
7. Your program must retrieve and display the stored data from the respective objects to the user.
8. Your program must properly calculate the Boss's lifespan.
9. Your Member ID must appear as a comment at the top of the main source code file.
10. Your program must be commented appropriately.

Example Input/Output:

```
Hero Data Entry:
Enter the hero name: Knight
Enter the character ID number: 111
Enter the hero level: 1
Enter the hero loot value: 50

Boss Data Entry:
Enter the boss name: Ogre
Enter the character ID number: 333
Enter the boss level: 2
Enter the boss hp: 5500
Enter the boss attack damage: 99

Hero information:
Name: Knight
ID number: 111
Level: 1
Loot: $50.00

Boss information:
Name: Ogre
ID number: 333
Level: 2
HP: 5500.0
Attack Damage: 99.0
Lifespan: 18.33 attacks
```

Solution and Project

The project is present on the flash drive _____ 10 points

The project contains employee.py and employeeinformation.py _____ 10 points

Program Execution

The program runs/compiles from the USB flash drive _____ 10 points

If the program does not execute, then the remaining items in this section receive a score of zero.

The program prompts the user to enter in the correct ProductionWorker attributes for Employee A _____ 10 points

The program prompts the user to enter in the correct ShiftSupervisor attributes for Supervisor A _____ 10 points

The program properly retrieves and displays the Employee A data. _____ 30 points

The program properly retrieves and displays the Supervisor A data. _____ 30 points

The program properly calculates Supervisor A's production bonus. _____ 50 points

Data is properly displayed (ie. formatted currency with two decimals points). _____ 20 points

Source Code Review

A comment containing the Member ID is present _____ 10 points

Functions and code sections are commented _____ 20 points

Employee class is created with appropriate accessor and mutator methods _____ 50 points

ProductionWorker class is created with appropriate accessor and mutator methods _____ 50 points

ShiftSupervisor class is created with appropriate accessor and mutator methods _____ 50 points

Input and output code is properly formatted. _____ 30 points

Code uses a consistent variable naming convention _____ 10 points

Total Points = _____ / 400 points